

Yupeng Su

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Research Interests

Efficient Pre-training / Post-training Optimization for Large Language Models: focusing on quantization, sparsification, and low-rank decomposition techniques to improve training and inference efficiency.

High-Performance Computing Infrastructure on GPU / FPGA Architectures: including compiler optimization and GPU/FPGA kernel adaptation for high-performance, energy-efficient inference and practical edge deployment.

Education

University of California, Santa Barbara, PhD in Computer Science – CA, USA Sept 2025 – present

- Advisor: Prof. Zheng Zhang (ECE, UCSB). GPA: 4.0/4.0.

Southern University of Science and Technology, BE in Microelectronics – China Sept 2021 – July 2025

- Advisor: Prof. Hao Yu (SME, SUSTech). GPA: 3.93/4.0 (178 units completed). Awards: Nominee of Top Ten Graduates of SUSTech; Top Ten Graduates of the College of Engineering.
- Dissertation: *Enhanced Mix-Precision Post-Training Quantization for Large Language Models: Block Awareness and Outlier Identification*.

Publications

MuonQ: Enhancing Low-Bit Muon Quantization via Directional Fidelity Optimization July 2026

Yupeng Su, Ruijie Zhang, Ziyue Liu, Yequan Zhao, Zheng Zhang
arxiv.org/abs/2605.11396 (CoLM, 2026)

RankGuide: Tensor-Rank-Guided Routing and Steering for Efficient Reasoning July 2026

Jiayi Tian, Yupeng Su, Ryan Solgi, Souvik Kundu, Zheng Zhang
arxiv.org/abs/2604.16694 (CoLM, 2026)

Muon²: Boosting Muon via Adaptive Second-Moment Preconditioning Apr 2026

Ziyue Liu, Ruijie Zhang, Zhengyang Wang, Yequan Zhao, Yupeng Su, Zi Yang, Zheng Zhang
arxiv.org/abs/2604.09967 (Under Review)

MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training Feb 2026

Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Yupeng Su, Liyan Tan, Zheng Zhang
arxiv.org/abs/2602.21545 (Under Review)

TEON: Tensorized Orthonormalization Beyond Layer-Wise Muon for Large Language Model Pre-Training Jan 2026

Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Dongyang Li, Yupeng Su, Sijia Liu, Zheng Zhang
arxiv.org/abs/2601.23261 (Under Review)

APTQ+: Attention-FFN-aware Post-Training Quantization for a Layer-wise LLM Accelerator on FPGA May 2026

Ziyi Guan, Zhengfei Chen, Dongkuan Wu, Ao Shen, Yifan Guo, Yupeng Su, Graziano Chesi, Mingqiang Huang, Ngai Wong, Hao Yu
10.1109/TCAD.2026.3697679 (IEEE TCAD, 2026)

- LLM-Barber: Block-Aware Rebuilder for Sparsity Mask in One-Shot for Large Language Models** Nov 2025
Yupeng Su, Ziyi Guan, Xiaoqun Liu, Tianlai Jin, Dongkuan Wu, Zhengfei Chen, Graziano Chesi, Ngai Wong, Hao Yu
[10.1109/ICCAD66269.2025.11240704](https://arxiv.org/abs/10.1109/ICCAD66269.2025.11240704) (IEEE/ACM ICCAD, 2025)
- PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models** Sept 2025
 He Xiao, Runming Yang, Qingyao Yang, Wendong Xu, Zhen Li, **Yupeng Su**, Zhengwu Liu, Hongxia Yang, Ngai Wong
arxiv.org/abs/2509.16989 (Under Review)
- Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning** Jan 2025
 Zhen Li*, **Yupeng Su***, Songmiao Wang, Runming Yang, Congkai Xie, Aofan Liu, Ming Li, Jiannong Cao, Yuan Xie, Ngai Wong, Hongxia Yang
arxiv.org/abs/2501.03035 (Under Review (*Equal Contribution))
- EdgeLLM: A Highly Efficient CPU-FPGA Heterogeneous Edge Accelerator for Large Language Models** May 2025
 Mingqiang Huang, Ao Shen, Kai Li, Haoxiang Peng, Boyu Li, **Yupeng Su**, Hao Yu
[10.1109/TCSL.2025.3546256](https://arxiv.org/abs/10.1109/TCSL.2025.3546256) (IEEE TCAS I: Regular Papers, 2025)
- APTQ: Attention-aware post-training mixed-precision quantization for large language models** June 2024
 Ziyi Guan, Hantao Huang, **Yupeng Su**, Hong Huang, Ngai Wong, Hao Yu
[10.1145/3649329.3658498](https://arxiv.org/abs/10.1145/3649329.3658498) (IEEE/ACM DAC, 2024)

Experience

- Student Research Assistant**, Prof. Hongxia Yang's Lab @ PolyU – Hong Kong, China Nov 2024 – May 2025
- Investigated the effects of compression on reasoning and explored data-driven strategies for restoration.
- Student Research Assistant**, Next Gen AI (NGai) Lab @ HKU – Hong Kong, China Aug 2024 – Feb 2025
- Contributed to lightweight networks for efficient inference under specified computing architectures.
- Summer Intern**, High Performance Integrated Circuit Design Lab @ SUSTech – China June 2024 – Sept 2024
- Implemented end-to-end pipelines for compressing, optimizing, and deploying models for efficient devices.
 - Published **EdgeLLM** based on the project and its results.